Energy, Work & Power

ANSWER KEY

Work done by a constant force

1 A force of 40 N pushes a crate a distance of 6 meters along a horizontal floor, in the direction of motion. Find the work done.

$W = Fd = 40(6) = 240$ J.

2 A rope pulls a sled with a force of 80 N at an angle of 30° above the horizontal, moving it 10 meters horizontally. Find the work done.

$W = Fd\cos\theta = 80(10)\cos(30^\circ) = 800(0.8660) \approx 692.8$ J.

3 A force of 25 N pulls a box a distance of 12 meters along a horizontal floor, in the direction of motion. Find the work done.

$W = Fd = 25(12) = 300$ J.

Kinetic energy

4 Find the kinetic energy of a car of mass 1000 kg traveling at 25 m/s.

$KE = \frac{1}{2}mv^2 = \frac{1}{2}(1000)(625) = 312500\text{ J} = 312.5$ kJ.

5 Find the kinetic energy of a cyclist and bicycle with combined mass 90 kg traveling at 8 m/s.

$KE = \frac{1}{2}mv^2 = \frac{1}{2}(90)(64) = 2880$ J.

6 Find the kinetic energy of a ball of mass 0.2 kg traveling at 15 m/s.

$KE = \frac{1}{2}mv^2 = \frac{1}{2}(0.2)(225) = 22.5$ J.

Gravitational potential energy

7 Find the gravitational potential energy gained by a 60 kg hiker climbing 150 meters, using $g = 9.8$ m/s^2.

$PE = mgh = 60(9.8)(150) = 88200\text{ J} = 88.2$ kJ.

8 Find the gravitational potential energy gained by a 5 kg object lifted 12 meters, using $g = 9.8$ m/s^2.

$PE = mgh = 5(9.8)(12) = 588$ J.

9 Find the gravitational potential energy gained by a 15 kg object lifted 8 meters, using $g = 9.8$ m/s^2.

$PE = mgh = 15(9.8)(8) = 1176$ J.

The work-energy theorem

10 A ball of mass 0.5 kg is moving at 4 m/s when a resultant force acts on it, increasing its speed to 10 m/s. Find the work done on the ball, using the work-energy theorem.
$W = \Delta KE = \frac{1}{2}(0.5)(10^2) - \frac{1}{2}(0.5)(4^2) = 25 - 4 = 21$ J.

11 A car of mass 900 kg is moving at 20 m/s when the brakes bring it to rest. Find the work done by the braking force, using the work-energy theorem.
$W = \Delta KE = 0 - \frac{1}{2}(900)(20^2) = 0 - 180000 = -180000$ J (negative, since the braking force removes kinetic energy).

12 A skateboard and rider of mass 60 kg is moving at 3 m/s when a force accelerates them to 9 m/s. Find the work done, using the work-energy theorem.
$W = \Delta KE = \frac{1}{2}(60)(9^2) - \frac{1}{2}(60)(3^2) = 2430 - 270 = 2160$ J.

Conservation of mechanical energy

13 A ball of mass 2 kg is released from rest at the top of a smooth (frictionless) ramp of height 5 meters. Find its speed at the bottom, using conservation of energy and $g = 9.8$ m/s^2.
PE lost $= KE$ gained: $mgh = \frac{1}{2}mv^2$, so $v^2 = 2gh = 2(9.8)(5) = 98$, giving $v = \sqrt{98} \approx 9.9$ m/s.

14 A pendulum bob of mass 1 kg swings from a height of 0.8 meters above its lowest point. Find its speed at the lowest point, using $g = 9.8$ m/s^2.
$v^2 = 2gh = 2(9.8)(0.8) = 15.68$, so $v = \sqrt{15.68} \approx 3.96$ m/s.

15 A skateboarder of mass 50 kg starts from rest at the top of a smooth (frictionless) ramp of height 3.2 meters. Find their speed at the bottom, using conservation of energy and $g = 9.8$ m/s^2.
$v^2 = 2gh = 2(9.8)(3.2) = 62.72$, so $v = \sqrt{62.72} \approx 7.92$ m/s.

Energy loss due to friction or resistance

16 A skier of mass 70 kg slides down a slope, descending a height of 40 meters. Their speed at the bottom is measured at 20 m/s. Find the energy lost to friction and air resistance, using $g = 9.8$ m/s^2.
PE lost $= mgh = 70(9.8)(40) = 27440$ J. KE gained $= \frac{1}{2}(70)(20^2) = 14000$ J. Energy lost to resistance $= 27440 - 14000 = 13440$ J.

17 A sled of mass 20 kg slides down a slope, descending a height of 10 meters. Its speed at the bottom is measured at 12 m/s. Find the energy lost to friction, using $g = 9.8$ m/s^2.
PE lost $= mgh = 20(9.8)(10) = 1960$ J. KE gained $= \frac{1}{2}(20)(12^2) = 1440$ J. Energy lost to friction $= 1960 - 1440 = 520$ J.

18 A crate of mass 30 kg slides down a ramp, descending a height of 6 meters. Its speed at the bottom is measured at 8 m/s. Find the energy lost to friction, using $g = 9.8$ m/s^2.
PE lost $= mgh = 30(9.8)(6) = 1764$ J. KE gained $= \frac{1}{2}(30)(8^2) = 960$ J. Energy lost to friction $= 1764 - 960 = 804$ J.

Power

19 A motor does 18000 J of work in 12 seconds. Find its power output.
$P = \frac{W}{t} = \frac{18000}{12} = 1500$ W.

20 A car engine produces a constant driving force of 2000 N while traveling at a constant speed of 18 m/s. Find the power output, using $P = Fv$.

$P = Fv = 2000(18) = 36000$ W $= 36$ kW.

21 A pump does 54000 J of work in 45 seconds. Find its power output.
$P = \frac{W}{t} = \frac{54000}{45} = 1200$ W.

22 A crane does 63000 J of work in 70 seconds. Find its power output.
$P = \frac{W}{t} = \frac{63000}{70} = 900$ W.

A well-designed word problem: a cyclist climbing a hill

23 This question has two parts. A cyclist and bicycle have a combined mass of 80 kg. The cyclist rides up a hill, climbing a vertical height of 30 meters over a time of 60 seconds, starting and ending at the same speed of 6 m/s (so kinetic energy does not change). Air and rolling resistance do a total of 900 J of negative work against the cyclist during the climb. Use $g = 9.8$ m/s^2. (a) Find the total work the cyclist must do against gravity and resistance combined. (b) Find the cyclist's average power output during the climb.

(a) Work against gravity: $mgh = 80(9.8)(30) = 23520$ J. Total work needed (gravity plus overcoming resistance): $23520 + 900 = 24420$ J. (b) Average power $= \frac{W}{t} = \frac{24420}{60} = 407$ W.